

A Fail-Closed Certificate Architecture for Literal Gaussian-Prime-Ideal Ownership in a Level-(3) Bianchi Flow

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Abstract

This article formulates a fail-closed certificate architecture for asking whether primitive loxodromic owners in a fixed level-(3) Gaussian Bianchi geodesic flow can receive one literal Gaussian prime ideal. It is a review-adjudicated synthesis of a frozen 22-source corpus, rather than a theorem of ownership, a mechanism implementation, a quotient computation, or a finite-refinement experiment. The corpus identifies the Picard/Bianchi object, primitive and inversion-aware terminology, adjacent arithmetic-group algorithms, and exact ideal-arithmetic components. It does not provide either the project-specific owner law or the complete primitive-root, conjugacy, inversion, and canonical-unoriented quotient required for evaluation. The literal one-ideal codomain is therefore treated as a deliberately strict stress test, not as a canonical or literature-forced output. We separate two open interfaces: Gate M for mechanism admissibility and Gate Q for quotient completeness. Their independence means only that neither gate entails the other at the design level. Downstream collision performance is firewalled from selecting or repairing either interface. The resulting prospective protocol specifies typed serializers, positive and negative certificates, immutable mechanism registration, controlled stop states, and independent replay. Across the 22 inherited citation contexts, 13 exact passage locators are finalized and nine uses are explicitly limited to metadata-only records because no passage-bearing locator was finalized; each bound supports only its stated contextual use. No owner, quotient, statistic, or route advancement is reported.

Keywords: Bianchi geodesic flow; Gaussian prime ideals; primitive conjugacy classes; certificate architecture; strict codomain; evidence synthesis

Traditional Chinese Abstract / 繁體中文摘要

本文提出一套失敗即封閉的證書架構，用以檢驗固定層級三之高斯 Bianchi 測地流中，原始雙曲所有者是否能被一致地指派一個字面上的高斯素理想。研究僅綜整已凍結的二十二項來源及既有評審紀錄，不執行新定理證明、機制建置、商集計算或有限精化實驗。文獻足以界定 Picard/Bianchi 幾何對象、原始與反向辨識語彙、相鄰算術群演算法及理想算術工具，卻未提供本計畫專屬的所有者法則，也未完成最大根、共軛、取逆及無向正規化所需的完整商集。故單一字面素理想被明確視為刻意嚴格的壓力測試，而非典範、伽羅瓦穩定或由文獻必然導出的值域。本文把機制可容許性與商集完備性分成兩道仍開放的關卡；兩者僅在設計蘊涵上互不替代，下游碰撞分離表現不得用來選擇或修補任一關卡。架構進一步規定型別化序列化、正反證書、不可變註冊、停止狀態與獨立重播。所有引文均缺段落定位，故支持狀態仍為不確定；本文不宣稱任何所有者、商集、統計值或路線晉級。

關鍵詞： Bianchi 測地流；高斯素理想；原始共軛類；證書架構；嚴格值域；證據綜整

1 Introduction, Research Question, and Contribution Boundary

The study fixes one mathematical frame before performance is visible. Its dynamical object is the inherited unit-speed geodesic flow on a torsion-free level-(3) Gaussian Bianchi manifold. Hyperbolic arclength is the clock. A prospective owner is a primitive loxodromic conjugacy class; inversion identifies the primary unoriented owner, and powers are repetitions rather than new owners. The requested arithmetic output is one literal nonzero prime ideal of $\mathbb{Z}[i]$. Associates denote the same ideal, while conjugate prime ideals above a split rational prime remain distinct. These are frozen design definitions, not findings extracted from the corpus.

The research question is deliberately conditional: under that strict output type, what must be certified before a performance-independently registered owner mechanism could be evaluated on a certified finite owner population? The question does not assume that a lawful mechanism exists. It also does not ask whether a norm, rational prime, composite ideal, residue class, or unordered conjugate pair might be a useful alternative object. Those alternatives are relevant to the interpretation of the frame, but substituting one of them after review would answer a different question.

The central distinction is between two pre-performance obligations. The mechanism-admissibility obligation asks whether a frozen formula maps each admissible primitive unoriented owner to one literal Gaussian prime ideal while respecting conjugacy, inversion, and repetition. The quotient-completeness obligation asks whether the inherited finite matrix population has been converted into exact primitive roots and canonical unoriented classes with complete positive and negative certificates. They are design-level independent: a formula cannot decide the quotient merely by producing labels, and a complete quotient cannot invent or validate a formula. This is not a proposition that the two failure events are statistically independent, nor a theorem that every possible construction separates into these modules.

The article's contribution is accordingly procedural and falsifiable. It specifies a fail-closed sequence in which neither classification performance nor collision separation can select a mechanism, repair an invalid codomain map, or excuse an incomplete owner quotient. A mechanism can fail; a quotient can remain unresolved; the downstream estimand can remain not evaluable. These are valid outcomes. The architecture is not claimed to be novel, and it is not presented as an existence or nonexistence theorem about intrinsic arithmetic ownership.

1.1 Claim classes and failure semantics

Four kinds of statements must remain distinct throughout the article. First, the frozen object definitions specify what this study elects to mean by flow, clock, primitive owner, repetition, inversion, and arithmetic output. They make the question determinate, but they do not show that an owner mechanism exists. Second, corpus findings describe only what the bounded evidence package can support at its present resolution. A source may identify the relevant group, discuss primitive geodesics, or provide an adjacent algorithm without instantiating the map required here. Third, architectural requirements state what a future implementation would have to serialize and verify. They are normative proof obligations, not completed constructions. Fourth, status statements record what has actually happened: no mechanism has been admitted, no quotient has been certified, and no downstream statistic is available.

This classification prevents several common but consequential substitutions. A well-defined ideal factorization algorithm does not select an ideal for a geometric owner. A conjugacy routine for a neighboring input class is not automatically a solver for the frozen level-(3) relation. An invariant observed on matrix rows does not turn those rows into primitive unoriented owners. Conversely, failure to find a mechanism in a bounded search would not prove nonexistence. Each statement must be evaluated against its own input type, output type, witness, and quantifier.

The literal single-ideal codomain is kept fixed precisely because it is a stress test. The paper asks

whether an unoriented primitive owner contains enough representative-independent information to select one literal Gaussian prime ideal, with associates identified but conjugate split primes still distinct. A later choice to return a norm, rational prime, composite ideal, residue class, or unordered conjugate pair may define an interesting study, but it would alter the frozen output type. Such a change cannot be treated as a repair after the present mechanism or performance has been inspected. Keeping the strict target does not elevate it to a canonical one; it preserves the falsifiability of this particular question.

Fail-closed semantics give informative meaning to an incomplete pipeline. Missing data do not default to a negative theorem, and an unresolved relation does not default to inequality. Gate M can remain unevaluable because no formula has been frozen, or it can refute a particular frozen formula by a reproducible law violation. Gate Q can remain unevaluable because its exact decision procedures are absent, or stop unresolved under a prospectively fixed resource limit. These outcomes are more precise than a generic failure label. They identify which interface lacks evidence without granting conclusions about other mechanisms, broader codomains, or the global arithmetic of closed geodesics.

Reader map. Every edge in this map is prospective. `ObjectLedger/v1` validates a serialized projective matrix, subgroup evidence, loxodromic eligibility, and dependency hashes; a conforming eligible row yields `OBJECT_INPUT_VALIDATED`, while malformed input yields `OBJECT_ROW_MALFORMED` and ineligibility yields `OBJECT_ROW_INELIGIBLE_EXCLUDED`. Gate Q may map validated inputs to oriented primitive classes and then canonical unoriented owners only after complete root, conjugacy, inversion, and reconciliation decisions; otherwise it emits the quotient-not-certified outcome `QUOTIENT_NOT_EVALUABLE` or the resource-bounded `QUOTIENT_UNRESOLVED_STOP`. Gate M consumes only certified owners and one immutable candidate record and emits exactly `MECHANISM_ADMISSIBLE`, `SPLIT_IDEAL_CODOMAIN_OBSTRUCTION`, or `FORMAL_MAP_REFUTED` under the declared precedence. `PerformanceLedger/v1` may then emit a schema-valid empty/result skeleton, or stop as `PERFORMANCE_POPULATION_HASH_MISMATCH_STOP`, `PERFORMANCE_MECHANISM_HASH_MISMATCH_STOP`, `PERFORMANCE_POST_FREEZE_MUTATION_STOP`, or `PERFORMANCE_POPULATION_RECONCILIATION_STOP`. `IndependentReplayReceipt/v1` emits `REPLAY_PASS` only after every expected disposition and dependency hash agrees; otherwise it emits `REPLAY_FIRST_MISMATCH_STOP` with the first mismatch. Any missing or stopped node prohibits every downstream use. No mapped output has been observed.

2 Frozen Literature and Theoretical Frame

2.1 The Picard/Bianchi object does not itself supply an owner law

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt identifies $\mathrm{PSL}(2, \mathbb{Z}[i])$ as the Picard group, places it in three-dimensional hyperbolic space, and begins to describe a prime-geodesic study. The Round-3 source-finalization record binds that statement to *arXiv 1804.00275v3 abstract, opening 20 words* and to support-excerpt SHA-256 d98212a0a6ba0e5e2caf c6419550379f9b5598242c93379450c40228124ce7bc. No broader proposition from the earlier context sentence is attributed to this passage. This citation may not be transferred to a class-to-ideal construction or owner law. [3].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt identifies the Picard manifold as a quotient by $\mathrm{PSL}(2, \mathbb{Z}[i])$ and frames the work around the prime geodesic theorem. The Round-3 source-finalization record binds that statement to *arXiv 1903.05111v4 abstract, opening 20 words* and to support-excerpt SHA-256 87768914857566d1899833f1d8efeb5a0334024d f47bdbaee24cf74585ca5451. No broader proposition from the earlier context sentence is attributed to this passage. The source is not used to define the project's owner map. [14].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt says that a result of Bykovskii is generalized to the Gaussian integers and that an asymptotic formula is proved for the prime geodesic theorem. The Round-3 source-finalization record binds that statement to *arXiv 1911.01800v2 abstract, opening 20 words* and to support-excerpt SHA-256 73996d529a60a5d70ef597ba957555fc8c983f40a8a778a8ce4735f091082e52. No broader proposition from the earlier context sentence is attributed to this passage. Shared arithmetic vocabulary is not evidence for a unique literal-ideal assignment. [4].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt reports prime-geodesic theorems counting primitive closed geodesics on a compact hyperbolic three-manifold and mentions length and holonomy constraints. The Round-3 source-finalization record binds that statement to *publisher-deposited Crossref abstract, opening 20 words* and to support-excerpt SHA-256 1908fc5f58fa44df694aacd4bfed4ba31ba81592472fcec287cbab1e57d72870. No broader proposition from the earlier context sentence is attributed to this passage. No decision procedure is transferred to the frozen noncompact level-(3) relation. [8].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. The citation supplies neither an owner law nor a literal-ideal selection mechanism. [15].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. No project mechanism or owner assignment is inferred. [18].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt reports a Selberg-zeta approach to reducing an error-term exponent in a prime geodesic theorem. The Round-3 source-finalization record binds that statement to *arXiv 1705.05626v1 abstract, opening 20 words* and to support-excerpt SHA-256 69dc856a15c0a837211dd6f81fbface9c7f2116339ddaf362376078080f76d3c. No broader proposition from the earlier context sentence is attributed to this passage. The uncorrected record may not be transferred to any affected premise or owner mechanism. [1].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt states that an error-term exponent is corrected for a prime geodesic theorem on hyperbolic three-manifolds and in Park's theorem. The Round-3 source-finalization record binds that statement to *publisher-deposited Crossref abstract, opening 20 words* and to support-excerpt SHA-256 207daa6f0171d89e9028d40b4c33e28b5fcfd65268b495d8c82feb2a655177bb. No broader proposition from the earlier context sentence is attributed to this passage. The correction is not treated as a mechanism source or retraction-clearance record. [2].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt states that the work improves an error term in the prime geodesic theorem for the Picard manifold. The Round-3 source-finalization record binds that statement to *arXiv 2407.17959v3 abstract, opening 20 words* and to support-excerpt SHA-256 87640051cc332c4ee9226eec7d22751cdbf9661c044867aa917c4a0052546510. No broader proposition from the earlier context sentence is attributed to this passage. No transfer to the mechanism or quotient interfaces. [21].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. No direct owner formula is imported. [17].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition

from the earlier context sentence is attributed to this source at passage level. The setting is not evidence that a literal ideal owner is canonical. [10].

Taken together, these records support the registered distinction between knowing the object and possessing the project-specific owner law.

2.2 Quotient completeness is a composite certificate target

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt identifies arithmetic Kleinian groups as arithmetic lattices in $\mathrm{PSL}(2, \mathbb{C})$ and announces an algorithm taking such a group as input; the retained fragment does not state the algorithm's output. The Round-3 source-finalization record binds that statement to *arXiv 1206.0087v3 abstract, opening 20 words* and to support-excerpt SHA-256 86367a1da1f091194f48f78e6fc502579bb4b720c8796a408cf3d789191e3fc9. No broader proposition from the earlier context sentence is attributed to this passage. No implemented level-(3) primitive-unoriented quotient is transferred. [19].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. No serialization rule is imported. [12].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. No instantiated level-(3) solver is claimed. [13].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. Canonicalization certificates remain separate obligations. [22].

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. The algorithm is not transferred. [20].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt states that the conjugacy problem in word-hyperbolic groups is solvable in linear time. The Round-3 source-finalization record binds that statement to *publisher-deposited Crossref abstract, opening 20 words* and to support-excerpt SHA-256 343e3b409e2524a42da0b9163b60e3a0fb79fa09c7a53d006ab826edff53413c. No broader proposition from the earlier context sentence is attributed to this passage. No project certificate follows. [11].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt announces an algorithm that decides whether two matrices in $\mathrm{GL}(n, \mathbb{Q})$ are conjugate in $\mathrm{GL}(n, \mathbb{Z})$. The Round-3 source-finalization record binds that statement to *arXiv 1811.06190v2 abstract, opening 20 words* and to support-excerpt SHA-256 7e90ca690c3a5c390f6b5b38a4bd53af3aa3047f71accd17e91b931da3aa6728. No broader proposition from the earlier context sentence is attributed to this passage. Ambient equality or conjugacy in $\mathrm{GL}(n, \mathbb{Z})$ is not transferred to the level-(3) projective subgroup relation. [9].

The quotient target therefore remains a composition problem. It requires an exact projective-lift convention, loxodromic eligibility, maximal group-element roots, level-(3) conjugacy, inversion links, deterministic canonical unoriented identifiers, transitivity checks, and population accounting. A bounded search that does not find a conjugator is not a negative certificate. The corpus supplies possible ingredients but no one frozen source supplies this complete target.

2.3 Ideal arithmetic validates outputs but does not choose owners

Within the frozen corpus, this citation is retained only as an explicitly metadata-only source record. The bounded Round-3 finalization pass yielded no passage-bearing locator, so no substantive proposition from the earlier context sentence is attributed to this source at passage level. The algorithms do not select an ideal owner. [7].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt describes the paper as a discussion of basic problems in algorithmic algebraic number theory. The Round-3 source-finalization record binds that statement to *arXiv math/9204234v1 abstract, opening 20 words* and to support-excerpt SHA-256 507f3d1dd12f77f3fd1e8b13554ebb44cf2e0bf7532c7bd20c40be5a5056d074. No broader proposition from the earlier context sentence is attributed to this passage. No class-to-ideal selection law is imported. [16].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt describes practical algorithms in computational algebraic number theory and mentions applications to class field theory. The Round-3 source-finalization record binds that statement to *publisher-deposited Crossref abstract, opening 20 words* and to support-excerpt SHA-256 52ce18dc3fa8c0230703a4762072d4eea58baef28bd4281354d3e2bb82ffa4f0. No broader proposition from the earlier context sentence is attributed to this passage. The interfaces do not justify assigning one branch of a split prime to an unoriented owner. [5].

Within the frozen corpus, this citation is limited to the following source-side statement: the excerpt announces an algorithm taking an integral ideal in a number field as input; the retained fragment does not state the algorithm's output. The Round-3 source-finalization record binds that statement to *publisher-deposited Crossref abstract, opening 20 words* and to support-excerpt SHA-256 f6696fcdfc700a3bd877cf63873a112ab4070c49e8031a8dab953320ec374554. No broader proposition from the earlier context sentence is attributed to this passage. Ideal powers are not transferred to maximal roots of group elements or geodesic repetition. [6].

The literal codomain is a deliberately strict frozen frame, not a canonical or literature-forced target. For a registered candidate M on eligible representatives, Gate M requires the exact laws $M(h\gamma h^{-1}) = M(\gamma)$ for the frozen subgroup relation and $M(\gamma^{-1}) = M(\gamma)$ for descent to an unoriented owner. Gaussian conjugation $\mathfrak{p} \mapsto \bar{\mathfrak{p}}$ is a separate codomain involution. Thus a candidate that instead yields $M(\gamma^{-1}) = \overline{M(\gamma)}$ fails unoriented descent when the two split prime ideals are distinct. Such a failure refutes only that registered candidate under the literal single-ideal codomain; it does not rule out other mechanisms or broader codomains.

3 Executed Methodology: Frozen Evidence Synthesis Only

The historical evidence synthesis remains bounded to 48 deliberately inspected record manifestations: 12 duplicate manifestations were removed, 36 unique records entered title-or-abstract screening, 27 entered detailed metadata/scope screening, and 22 were admitted, of which 17 entered the peer-reviewed journal-or-correction numerator. Those historical aggregates are unchanged. To address only the disclosed replay gap, all 53 frozen query lines were rerun verbatim on 2026-09-04 under a top-three connector-visible-result cap with no pagination. The new replay inspected 139 manifestations, retained 89 unique current work keys after removing 50 current duplicates, resolved 19 representatives to already admitted P29-S identifiers, excluded 70 representatives from the fixed 22-source crosswalk under the recorded replay-only, source-type, or frozen-scope dispositions, and preserved 5 no-visible-result queries as explicit unavailable rows. P29-S04, P29-S16, and P29-S20 did not surface inside this bounded window; that observation is neither an existence verdict nor a change to the admitted corpus. Every current row carries retrieval, deduplication, and screening fields. The original-session screened-out row identifiers and

decisions remain unavailable and are not reconstructed from the new observations.

Each admitted record was encoded in a source-effect matrix with support class, claim-fitness grade, admissible use, prohibited stronger use, compatibility role, and locator limitation. The matrix was synthesized into the research report and then assessed from procedurally role-separated, same-model-family editorial, domain, methodology, and adversarial perspectives before author adjudication. Role separation does not remove correlated-error risk, and no model-family or review independence is claimed. This reader-facing account concerns evidence selection and synthesis only. It retrieved or created no scientific artifact, experiment, or result and changes no review disposition.

The method has an explicit evidentiary ceiling. In this paper, the 22 inherited prose citation contexts now split into 13 exact, row-bounded passage locators and nine explicit bounded-unavailability, metadata-only records. The exact locators identify an abstract, theorem-adjacent section, page, or paragraph surface and bind a short support excerpt; they do not discharge project-specific hypotheses or justify a stronger transfer. The unavailable rows do not guess a locator. Across the five-paper Round-10 batch, historical aggregate statements are retained only as workflow context and are not recalculated here. No direct quotation appears, and no scientific value, owner law, quotient result, or Route status is inferred.

Correction, source-status, and integrity checks remain separately typed. P29-S06 stays bound to P29-S07 for affected use. P29-S09 remains a preprint. Structured retraction status is NOT_CHECKED for all 22 source rows, and source-level conflict-of-interest status is UNKNOWN_NOT_AUDITED. The absence of a fabricated or unverifiable inventory row is not retraction clearance, and a correction binding is not a full theorem-applicability audit.

3.1 Transformation from evidence records to manuscript claims

Stage 2 converts the frozen evidence synthesis into a publication-shaped argument without increasing its scientific strength. Each registered claim has a preassigned source set and an explicit negative constraint. The manuscript may narrow a claim, explain its logic, or state a prospective certificate field, but it may not add an unassigned source, infer a stronger theorem, or turn an intended interface into an executed result. This rule is especially important where several adjacent sources appear to compose naturally: composition at the level of a research plan is not composition at the level of a proved terminating algorithm.

Source identity remains separate from passage support. The source inventory, DOI closure, bounded metadata checks, and support-role coding justify citing a record for a narrow contextual purpose. Thirteen row-specific exact locators now bound those contextual uses; nine other rows explicitly remain metadata-only because the bounded pass finalized no passage-bearing surface. Neither class identifies a project theorem or discharges its hypotheses. The correction relation between P29-S06 and P29-S07 is retained, and the preprint status of P29-S09 remains visible, but neither condition is converted into a general integrity clearance. This is a deliberate limit of the manuscript rather than a silent assumption.

The executed outcome of this transformation is therefore modest but concrete: a coherent argument, a closed bibliography with all 22 frozen identifiers, and an auditable set of claims whose prohibited upgrades remain legible. No numerical or algebraic computation is hidden inside the prose conversion. No candidate formula, matrix population, quotient ledger, collision table, or replay receipt was generated. The article can guide a later scientific stage, but it cannot serve as evidence that the later stage has already occurred.

4 Review-Adjudicated Proof and Method Architecture

4.1 Gate M: mechanism admissibility under the strict frame

Gate M is a prospective logical and serialization contract. Before any collision outcome is visible, one candidate mechanism must be frozen by immutable formula bytes, source trace, admissible domain, literal-ideal codomain, normalization, associate convention, and the equations $M(h\gamma h^{-1}) = M(\gamma)$ and $M(\gamma^{-1}) = M(\gamma)$ on the frozen owner relation. The validator must distinguish a nonzero Gaussian prime ideal from an element, generator, norm, rational prime, composite ideal, factorization list, residue, or unordered conjugate pair. It must also record whether Gaussian conjugation fixes or exchanges the output. A split branch must follow from the registered formula rather than from observed collisions, and failure is scoped to that exact formula.

Gate M has an explicit precedence rule. If the candidate violates literal split-branch compatibility under inversion, the primary disposition is `SPLIT_IDEAL_CODOMAIN_OBSTRUCTION` and the same receipt records `formal_map_refuted=true`. If that predicate passes but another declared formal law fails, the disposition is `FORMAL_MAP_REFUTED`. Only a candidate passing every declared law and output-type check may receive `MECHANISM_ADMISSIBLE`. These outcomes concern the exact registered candidate under the frozen frame, never all possible mechanisms or codomains, and none has been produced here.

4.2 Gate Q: primitive-unoriented quotient completeness

Gate Q is a prospective population-construction contract. Every eligible inherited matrix row would need an exact loxodromic decision, a maximal group-element root witness, a level-(3) conjugacy class assignment, an inversion link, and one canonical unoriented owner ID. Positive relations require reproducible witnesses. Negative relations require a complete decision procedure or an equivalently auditable certificate; exhaustion of a bounded search is not enough. Deterministic serializers must make repeated runs byte-comparable.

The gate remains unresolved independently of Gate M. If exact completeness cannot be established from the object and algorithms, the outcome is `QUOTIENT_NOT_EVALUABLE`. If a predeclared resource limit is reached while relations remain unresolved, the outcome is `QUOTIENT_UNRESOLVED_STOP`. A complete quotient is not inferred from the number of rows, apparent matrix uniqueness, or a mechanism's labels. No quotient computation or certificate exists in this phase.

4.3 Prospective certificate graph and replay obligations

The first prospective node is `ObjectLedger/v1`. Each row must contain canonical projective-matrix bytes, lift and central-sign conventions, subgroup-membership evidence, an exact loxodromic-eligibility disposition, serializer version, and hashes of every dependency. Its validator accepts only schema-conforming rows with reproducible membership and eligibility witnesses; malformed input yields `OBJECT_ROW_MALFORMED`, and an ineligible row yields an explicit exclusion receipt. Positive, negative, malformed, and serialization-round-trip fixtures must carry predeclared expected dispositions. No such ledger or fixture has been executed.

The second prospective node is `QuotientLedger/v1`. It must bind each admitted object hash to maximal-root, subgroup-conjugacy, inversion, transitive-closure, and canonical-owner decisions. Positive relations require replayable witnesses; negative decisions require a complete terminating procedure for the exact frozen relation. The validator rejects missing root exponents, unverifiable conjugators, inconsistent inverse links, nontransitive class IDs, and population totals that do not reconcile inputs, exclusions, repetitions, oriented classes, and unoriented owners. Root, inverse, conjugate, nonconjugate,

malformed-witness, and resource-limit fixtures must have frozen expected stop states. No quotient ledger or fixture has been executed.

The third prospective node is `MechanismRegistry/v1`. A record must bind immutable formula bytes, domain, codomain, source trace, normalization, associate convention, split-conjugate treatment, and the exact conjugacy, inversion, and repetition predicates. The validator separately checks literal prime-ideal type, representative independence, and every declared law, and it attaches any failure only to the record hash. Fixtures must include admissible inert outputs, split conjugate pairs, a deliberate inversion-law failure, a nonprime output, a malformed formula record, and a timeout or unavailable-dependency case, each with a predeclared typed disposition. No mechanism record or fixture has been executed.

The fourth prospective node is `PerformanceLedger/v1`. It may consume only the exact `QuotientLedger/v1` population hash and a `MECHANISM_ADMISSIBLE` registry hash. Its schema must freeze baseline collisions, holdout membership, controls, clock, and the definition of $S_H(M)$ before output access. The validator rejects raw-row scores, incomplete quotients, mismatched mechanism bytes, post-result control changes, or unreconciled populations. Fixtures must include a valid empty-result skeleton, a population-hash mismatch, a mechanism-hash mismatch, and a post-freeze mutation, all without supplying scientific values. No performance ledger or fixture has been executed.

The prospective `IndependentReplayReceipt/v1` must enumerate every consumed content hash, validator version, fixture identifier, expected disposition, observed disposition, and first mismatch. The verifier may reuse parsers and public schemas but must not import producer-side canonicalization, relation-decision, formula-evaluation, or scoring routines; those transformations require independently implemented checks or direct witness verification. Replay must cover positive, negative, malformed, inversion, root, and resource-limit fixtures and must fail closed on an unrecognized proof type. Agreement would certify only the registered finite procedure. No verifier, fixture execution, or receipt exists in the present work.

4.4 The limited meaning of independence

The two gates are independent in an architectural entailment sense only. Closure of Gate M does not entail closure of Gate Q because producing lawful labels does not decide primitive roots, conjugacy, or inversion classes. Closure of Gate Q does not entail closure of Gate M because a certified owner population does not define a literal ideal-selection law. Failure of either gate blocks downstream performance, but their error rates, computational costs, and empirical behavior have not been observed. The present article therefore makes no claim of mathematical, statistical, or empirical independence.

4.5 Performance firewall

Only after both gates pass could a prospectively frozen owner partition expose a finite collision-separation estimand such as $S_H(M)$. The statistic would count only predeclared baseline-collision owner pairs in a certified holdout for which a fixed mechanism returns different valid literal ideals. It could not choose the mechanism, repair a failed law, or compensate for duplicate/repeated owners. No value of S_H is reported. Even a positive future value would be a bounded finite-refinement result rather than a global orbit-to-prime-ideal theorem.

Typed controls would remain separate from the primary score. Permuting ideal labels without changing equivalence classes is a relabeling invariance test, not arithmetic specificity. Replacing literal primes with composite objects is a type violation, not a matched control. Directionally sensitive controls would need to be registered before output access and assessed under the same object, owner, and clock definitions.

Prospective control interpretation. Each control and stop below is unexecuted. An owner-label permutation preserves the quotient and tests scoring sensitivity to label identity; failure produces `CONTROL_LABEL_DEPENDENCE_STOP`, diagnoses label dependence, and cannot establish arithmetic specificity. An inversion-paired control preserves the underlying owner population while testing representative invariance; failure produces `CONTROL_REPRESENTATIVE_INVARIANCE_STOP`, diagnoses an owner-law or serialization defect, and cannot identify the correct ideal branch. A broadened-codomain control, such as an unordered conjugate pair, changes the output type; noncomparability produces `CONTROL_CODOMAIN_COMPARABILITY_STOP` and cannot rescue the registered single-ideal mechanism, count as performance under the frozen estimand, or establish an all-codomain obstruction. Every construction, statistic, tolerance, and expected disposition remains a pre-output registration obligation, and neither a control result nor any listed stop has been observed.

5 Evidence-Synthesis Findings

The first registered finding is a support boundary. The frozen corpus identifies the Picard/Bianchi geodesic object and supplies primitive, inversion, group-algorithm, and ideal-arithmetic vocabulary. It does not define the project-specific Gaussian-prime-ideal owner law. Object identity and terminology therefore cannot be promoted into an owner map, a quotient, or a refinement score.

The second finding is that quotient completeness remains a composite certificate target. The adjacent methods do not arrive as one theorem or implemented solver for the exact level-(3) relation. The required hypothesis bridges, termination logic, positive and negative witnesses, and deterministic canonicalization remain prospective. This gap is genuine within the frozen design even if the literal codomain were later replaced in a different study.

The third finding is the review-adjudicated codomain interpretation. One literal Gaussian prime ideal is a deliberately strict project frame, not an intrinsic consequence of the geometry or literature. A split-branch obstruction under that frame would be conditional and construct-specific. An unordered Galois orbit or another equivariant output might avoid that problem, but considering it would require a separately frozen research question; it cannot be introduced as a repair after performance access.

The fourth finding is the pre-performance separation of mechanism admissibility and quotient completeness. Their design-level non-entailment prevents performance from selecting or repairing either interface. The contribution is the resulting fail-closed certificate architecture, not a theorem, novelty determination, implemented mechanism, or scientific result.

6 Reproducibility and Prospective Implementation Interface

The four frozen manuscript-audit sources remain available at the commit-pinned repository locator https://github.com/maris205/hilbert-polya-structure/tree/5ca65027918c0fce7ef9af82f3faf2e46ed6530c/flow_systems/papers/29-bianchi-ideal-owner-refinement:notes/stage1_phase2_annotated_bibliography.md, SHA-256 c4d71637e5676337326d2eb78dcdd64d78b4b116a397c50c54a081d7c5e2650b; notes/stage1_phase2_source_inventory.tsv, 67ed7713bd6881d11466dc16755c7660a458c52e07ee072d086d6467f8ad7bd8; notes/stage1_phase2_source_verification.tsv, bcf5fa7af07f353fbcaaa6fca319e79f173d7a6af070b58276a91fe9a44d8901; and notes/stage1_phase3_literature_matrix.tsv, 219309f0c8cea9106ba162c95f2de266b78b1c5d8e0d541e5b4a0e8122247d2c. The 2026-09-04 run-local replay is separately bound as notes/stage4_prime_literature_replay_round2.raw.json (schema p29-dated-literature-replay/1.0, 3c7139a23df40b3f9315af74aa6fbacc7cbc06f7d349e9b4e785e45ad2ed28f3); its row ledger is notes/stage4_prime_literature_screenin

g_ledger_round2.json (schema p29-current-row-retrieval-dedupe-screen-ledger/1.0, a38a89f7802f6199ebe4de94217075abfe61b1707d7c0b730dd99989da2e0f81) with TSV projection notes/stage4_prime_literature_screening_ledger_round2.tsv (2d4bf2fb0c63f0c9904fcd16dab90296278499c4611c83214482950fd8c6ac2); and the 22-row admitted-source crosswalk is notes/stage4_prime_inventory_matrix_crosswalk_round2.json (schema p29-source-inventory-to-literature-matrix-crosswalk/1.0, fde57cac6ae3a41d9d895eff6a7a480b1b9f560bed4f271ec7aa97eda8929016) with TSV projection notes/stage4_prime_inventory_matrix_crosswalk_round2.tsv (c208570909fcf8b1e9902c5fac1471ce3a8392b04cddb91848cc933465300b9). These run-local files support only bounded replay provenance and deterministic P29-S01–P29-S22 row linkage; they do not refresh source status or scientific results.

The completed Round-3 source-finalization inputs are notes/stage4_prime_source_finalization_round3.json (SHA-256 05997bc748c453d01e9a5674528acfeb496ffdb9b8d7d6ef22c6e8d30c2bffd), the 22-row claim–passage matrix notes/stage4_prime_claim_passage_matrix_round3.json (SHA-256 ac253359ce62df4c4f7d8c1143fde92d71918c157c45a68f8a32717d3bc79b71), and its deterministic validation receipt notes/stage4_prime_source_finalization_round3_validation.json (SHA-256 9373e2bd962a9d865a1a3690b5b0cf17fd3d01336578c3fde665692c5bb4b26e). The matrix records exactly 13 finalized passage locators and nine explicit bounded-unavailability, metadata-only records. Each locator supports only its row’s stated contextual use; none supplies a project-specific owner law, quotient procedure, scientific result, or Route credit.

A later scientific implementation would expose five closed interfaces: ObjectLedger/v1, QuotientLedger/v1, MechanismRegistry/v1, PerformanceLedger/v1, and IndependentReplayReceipt/v1. Each interface must publish canonical bytes, dependency hashes, an allowed proof-type registry, validator acceptance predicates, and typed stop states. Producer and verifier may share only schemas, parsers, and witness formats; the verifier may not reuse producer-side canonicalization, relation-decision, mechanism-evaluation, or scoring logic. The required positive, negative, malformed, inversion, root, and resource-limit fixtures are specifications with expected dispositions, not executed tests. Accordingly, no replay, robustness, sensitivity, or usefulness result is reported.

All five interfaces are prospective. No serializer, engine, registry entry, control result, or replay receipt is claimed to exist at Stage 2. Their purpose is to define what evidence would be needed for later evaluation and what stop state must be returned when evidence is missing.

7 Discussion and Implications

Pending implementation, the strict literal codomain has prospective organizational stress-test value only: it forces the owner input, codomain, invariance law, and typed failure scope to be declared before output access. The labeled synthetic fixture SF-LITERAL-01 is defined, but not run, in notes/stage4_prime_sf_literal_01_definition_round2.json (schema p29-synthetic-fixture-definition/1.0, SHA-256 902f1acbee591692f8c5f533395900448911d66feda7904d5e56513f39804b3d). It freezes one synthetic unoriented-owner token with formal representatives γ_{sf} and γ_{sf}^{-1} , an unordered-conjugate-pair baseline that returns $\{p_{sf}, \bar{p}_{sf}\}$ for both representatives, and a fixture-local literal-branch candidate returning the two predeclared distinct branch tokens separately. The expected validator disposition is SPLIT_IDEAL_CODOMAIN_OBSTRUCTION with formal_map_refuted=true, scoped only to that fixture-local candidate under the single-ideal codomain. This expected disposition is a test oracle, not an observed control, performance value, owner computation, or obstruction for every broader codomain.

The quotient gate has a different status. It concerns the validity of the observational unit. Rows cannot become primitive owners through labeling, and finite performance on uncertified rows would not estimate the registered construct. This obligation persists under alternative codomains and is therefore independent of the split-frame question in the limited architectural sense defined above.

The combined architecture encourages interpretable negative outcomes. It distinguishes absence of a frozen admissible mechanism from formal refutation of one mechanism, and it distinguishes an unavailable exact quotient from a resource-bounded unresolved stop. Those distinctions prevent an engineering failure from being overread as a theorem and prevent a finite numerical success from concealing a type error.

The bounded contribution is a project-specific synthesis of typed owner semantics, a literal-ideal stress test, fail-closed Gate-Q and Gate-M contracts, and an independent-replay specification. Certificate methods, proof-carrying computation, and replay-oriented workflow design are therefore comparison classes rather than components claimed as new in isolation. The frozen source study did not perform a field-general novelty or priority review, so no global originality claim is made. Because the prospective schemas, fixtures, controls, and performance ledger have not been executed, practical usefulness and scientific performance also remain unevaluated.

8 Acknowledged Limitations

The source study remains bounded rather than exhaustive. The frozen files disclose interfaces, dates, exact queries, normalization and deduplication rules, admitted identifiers, and source-role decisions, but the original-session screened-out row identifiers and inclusion or exclusion decisions remain unavailable. The 2026-09-04 replay adds a complete row ledger only for its declared top-three, no-pagination observation window; it is time-dependent new evidence and does not reconstruct the missing historical rows. Five queries returned no visible row within that bound, and three frozen admitted identifiers did not resurface there. The original aggregates, the fixed 22-source corpus, and every source-status label remain unchanged. Neither the replay nor the deterministic admitted-source crosswalk supports literature-absence, exhaustive-coverage, novelty, or scientific-performance claims.

Of the 22 paper-specific citation contexts, 13 have exact, row-bounded passage locators and support-excerpt digests, while nine are explicit bounded-unavailability, metadata-only records. Statements near theorem boundaries remain narrowed to the recorded contextual scope. The locator work certifies no theorem-hypothesis transfer, maximal-root formula, conjugacy procedure, or split-ideal fact for this project. P29-S06/P29-S07 correction handling remains mandatory, structured retraction screening was not run, and source-level conflicts were not audited. Historical five-paper batch counts are workflow context, not a refreshed scientific or bibliographic result.

No separately authorized field-wide novelty analysis was performed. Passage adjudication is limited to 13 exact contextual locators and nine explicit metadata-only unavailability records in the 22-row matrix. The contribution therefore remains the manuscript's project-specific synthesis and prospective specification; it is not a priority claim against all certificate-methods, proof-carrying computation, or arithmetic-dynamics workflows. The exact locators do not strengthen any scientific claim, while unavailable rows authorize no passage transfer. The machine-audit comments preserve those boundaries.

No mechanism, quotient, owner population, collision set, S_H value, control result, or replay receipt exists here. No nonexistence theorem follows from the missing mechanism, and no universal split-prime obstruction is asserted. The literal codomain remains fixed for this study even though it is not claimed to be intrinsically privileged. The exact quotient remains independently unresolved.

The next authorization should also preserve a strict order of access. Passage adjudication should occur before a theorem is mapped to the object; the theorem-to-object map should be frozen before an implementation is evaluated; and both Gate M and Gate Q artifacts should be sealed before collision

labels or holdout performance are exposed. This order is not a claim that one implementation strategy is uniquely correct. It is a safeguard against choosing definitions, stopping rules, or split branches because they improve a visible finite score.

A later manuscript could report partial progress without weakening this order. For example, it could close source locators while leaving both scientific gates open, instantiate a complete quotient while reporting no admissible mechanism, or refute one registered mechanism while leaving the broader search unresolved. Each outcome would require its own hashes and receipts and would retain the strict distinction between failure of a candidate and nonexistence in the whole hypothesis class. Such modular reporting is preferable to a single binary success label because it keeps the mathematical scope of every negative conclusion visible.

9 Future Work

The smallest source-level step would be a separately authorized locator pass for the few premises closest to an owner formula, group-root decision, and literal split-branch law. Any exact passage would need its hypotheses and prohibited transfers frozen; identity verification alone is insufficient.

The smallest object-level step would be a theorem-to-object map for projective lifts, subgroup membership, loxodromic status, maximal roots, level-(3) conjugacy, inversion, and deterministic serialization. It should specify stop states before implementation. A mechanism search and a quotient implementation should remain separate until both are independently frozen.

Only after those steps should a scholar-confirmed scientific design register a mechanism, owner partition, holdout, controls, resource limits, and replay protocol. Alternative Galois-stable or oriented codomains may be studied later as distinct hypotheses, not as post hoc replacements for the frozen literal target.

10 Conclusion

The frozen literature is sufficient to formulate a strict, auditable ownership study but not to produce an owner law or a certified owner population. The present article makes the construct boundary explicit: the literal Gaussian-prime-ideal codomain is deliberately strict and frozen, not canonical or forced, so any split obstruction is conditional on that frame. Quotient completeness is a separate unresolved problem. The two gates are independent only as non-entailing design obligations, and performance is downstream of both.

The article’s defensible result is an evidence-synthesis and certificate architecture with explicit failure states and prospective interfaces. It is not a theorem, scientific computation, novelty finding, or Route advance. The formal Route-A tuple remains UNASSIGNED, positive arithmetic A2 remains 0/1, and Route B remains uninvoked.

The immediate value of the architecture is diagnostic: it prevents a missing owner law, an incomplete observational quotient, and weak finite separation from being collapsed into one ambiguous outcome. A later study can close, refute, or stop at each interface with evidence appropriate to that interface. Until such artifacts exist, Gate M and Gate Q remain open and $S_H(M)$ remains undefined rather than zero.

Author Contributions

Liang Wang is the responsible human author. CRediT-style roles: Conceptualization; Methodology; Writing—review and editing; Project administration. He supplied the conceptual direction, made the author-adjudication decisions, approved the recorded workflow gates, interpreted the bounded evidence

synthesis, and accepts final accountability for the manuscript. AI assistance is disclosed separately and is not credited with authorship.

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Competing Interests

The author declares no competing interests.

Ethics Statement

Ethics approval, informed consent, and human-subject review are not applicable. This article concerns mathematical objects, published-source records, and project-owned workflow artifacts; it reports no human participants, intervention, identifiable personal data, or human-related dataset.

Data and Materials Availability

The four frozen manuscript-audit files remain bound by the digests reported in the reproducibility section. Run-local support files add the 2026-09-04 raw query replay (notes/stage4_prime_literature_replay_round2.raw.json, schema p29-dated-literature-replay/1.0, SHA-256 3c7139a23df40b3f9315af74aa6fbacc7cbc06f7d349e9b4e785e45ad2ed28f3), the JSON/TSV current-row ledger (notes/stage4_prime_literature_screening_ledger_round2.json, a38a89f7802f6199ebe4de94217075abfe61b1707d7c0b730dd99989da2e0f81; notes/stage4_prime_literature_screening_ledger_round2.tsv, 2d4bf2fb0c63f0c9904fcd16dab90296278499c4611c83214482950fd8c6acb2) and the JSON/TSV P29-S01–P29-S22 crosswalk (notes/stage4_prime_inventory_matrix_crosswalk_round2.json, fde57cac6ae3a41d9d895eff6a7a480b1b9f560bed4f271ec7aa97eda8929016; notes/stage4_prime_inventory_matrix_crosswalk_round2.tsv, c208570909fcf8b1e9902c5fac1471ce3a8392b04cddb91848cc933465300b9). The ledger schema is p29-current-row-retrieval-dedupe-screen-ledger/1.0; the crosswalk schema is p29-source-inventory-to-literature-matrix-crosswalk/1.0. Historical screened-out rows and their original decisions remain unavailable. No owner mechanism, quotient ledger, certificate execution, control result, experiment, scientific value, or Route result was created.

The completed Round-3 source-finalization inputs are notes/stage4_prime_source_finalization_round3.json (SHA-256 05997bc748c453d01e9a5674528acfeb496ffdb9b8d7d6ef22c6e8d30c2bffdcc), the 22-row claim–passage matrix notes/stage4_prime_claim_passage_matrix_round3.json (SHA-256 ac253359ce62df4c4f7d8c1143fde92d71918c157c45a68f8a32717d3bc79b71), and its deterministic validation receipt notes/stage4_prime_source_finalization_round3_validation.json (SHA-256 9373e2bd962a9d865a1a3690b5b0cf17fd3d01336578c3fde665692c5bb4b26e). The matrix records exactly 13 finalized passage locators and nine explicit bounded-unavailability, metadata-only records. Each locator supports only its row’s stated contextual use; none supplies a project-specific owner law, quotient procedure, scientific result, or Route credit.

AI-Assistance Disclosure and Verification Limitation

OpenAI Codex, operating in the GPT-5 model family, assisted during sessions dated 2–4 September 2026 UTC; the exact backend snapshot or build was not exposed. Assistance covered organization of the frozen bibliography and source identities, bounded evidence synthesis, drafting and review integration, LaTeX conversion, deterministic checks of identifiers, citations, references, hashes, and word counts, and on 3–4 September the Stage-4/Stage-4-prime provenance work, bounded source finalization, scope-reissue preparation, authority projection, and writer-side patch emission. No AI-authored scientific data, experiment, mechanism, quotient, proof, or numerical result is presented. Procedural role separation inside one model family is not independent validation and retains a correlated-error limitation. AI assistance is not credited with authorship.

Liang Wang made the recorded stage-gate confirmations and author-adjudicated framing decisions. Neither that responsibility nor the present composition implies human full-text or source-passage verification. Stage 2.5 is complete. The bounded Round-3 source finalization records 13 exact passage locators and nine explicit metadata-only unavailability records across 22 contexts. Passage support is limited to those row-specific uses; the prior Stage-4.5 Round-1 verdict remains a correction-required failure until a fresh post-apply Stage-4.5 audit is separately authorized and passed. No owner, quotient, statistic, scientific result, or Route advancement follows from the locator work.

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